Ch 7 §8 Fourier Series

(we will cover Section 7 next $\Rightarrow$ solving $\Delta u = 0$ ~~on a~~ in polar coord on a finite domain).

Recall from Calc 2, we can represent certain fns in terms of Taylor series (as an ∞-long polynomial)

$$f(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{1}{2!} f''(x_0)(x-x_0)^2 + \cdots$$

ex) $\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$ $\quad (x_0 = 0)$

Q: What kind of fns have a Taylor series $\Rightarrow C^\infty(I)$, $x_0 \in I$

Q: Does the Taylor series converge to $f(x)$?

Not all C^∞ f'ns do: (not responsible for)

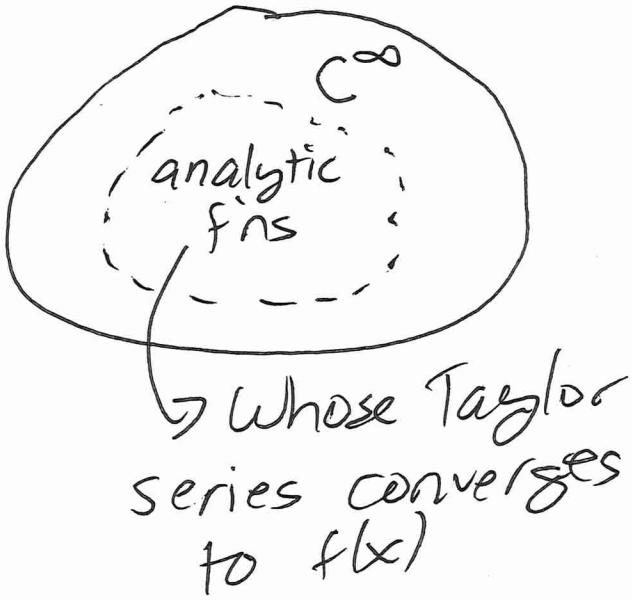
$$f(x) = \begin{cases} e^{-\frac{1}{x^2}} & x \neq 0 \\ 0 & x = 0 \end{cases}$$

$f(x) \in C^\infty$ (even at $x=0$)

but Taylor series about $x=0$

$f(0)=0 \quad f'(0)=0 \quad f''(0)=0 \ \ldots \ f^{(n)}(0)=0$

Taylor series for $f(x) = 0$



Fourier series

Instead of using $\{1, x, x^2, x^3, \dots\}$ as a "basis"

we use $\{1, \cos(x), \sin(x), \cos(2x), \sin(2x), \cos(3x), \sin(3x) \dots\}$.

$\boxed{\text{① These are all periodic f'ns.}}$ $0 \le \theta < 2\pi$ (one period)

$$f(\theta) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos(n\theta) + b_n \sin(n\theta) \right]$$

What do the coefficients mean?

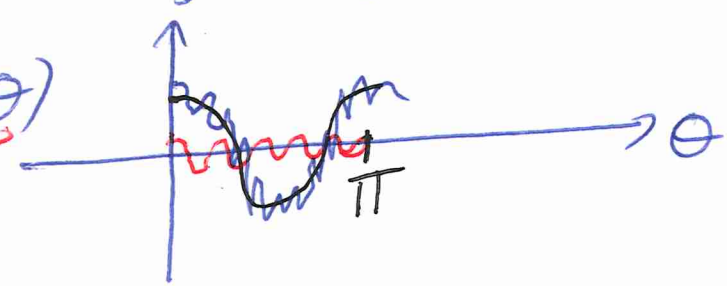
Let's suppose that all coeff are zero except for a_5

$$f(\theta) = a_5 \cos(5\theta)$$

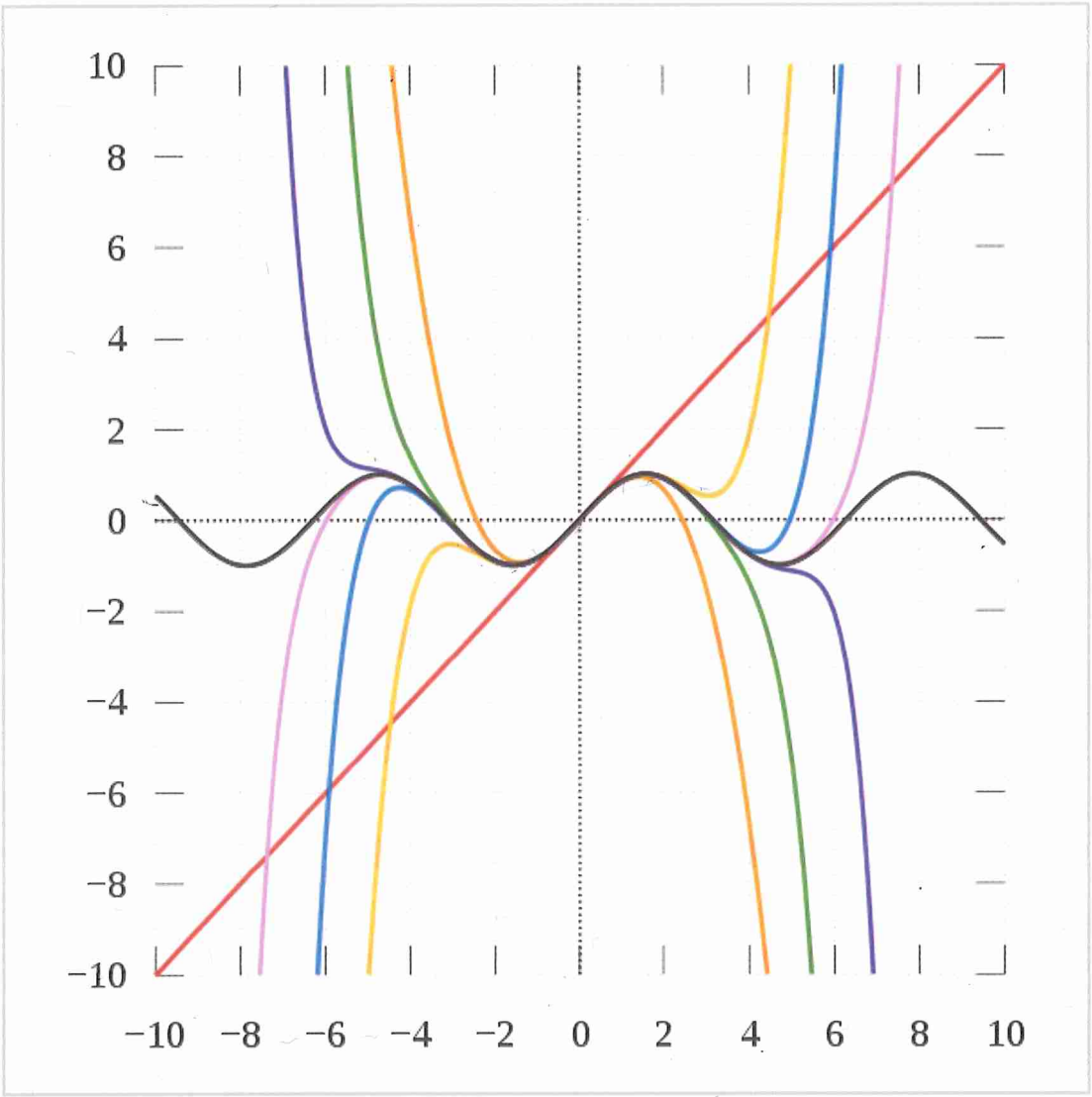
periodic. Frequency $f = \frac{1}{T} = \frac{5}{2\pi}$

T = period $= \frac{2\pi}{5}$ a_5 = magnitude of the freq.

$$\underbrace{f(\theta)}_{\text{blue}} = \underbrace{\cos(2\theta)}_{\text{black}} + \underbrace{2\cos(5\theta)}_{\text{red}}$$

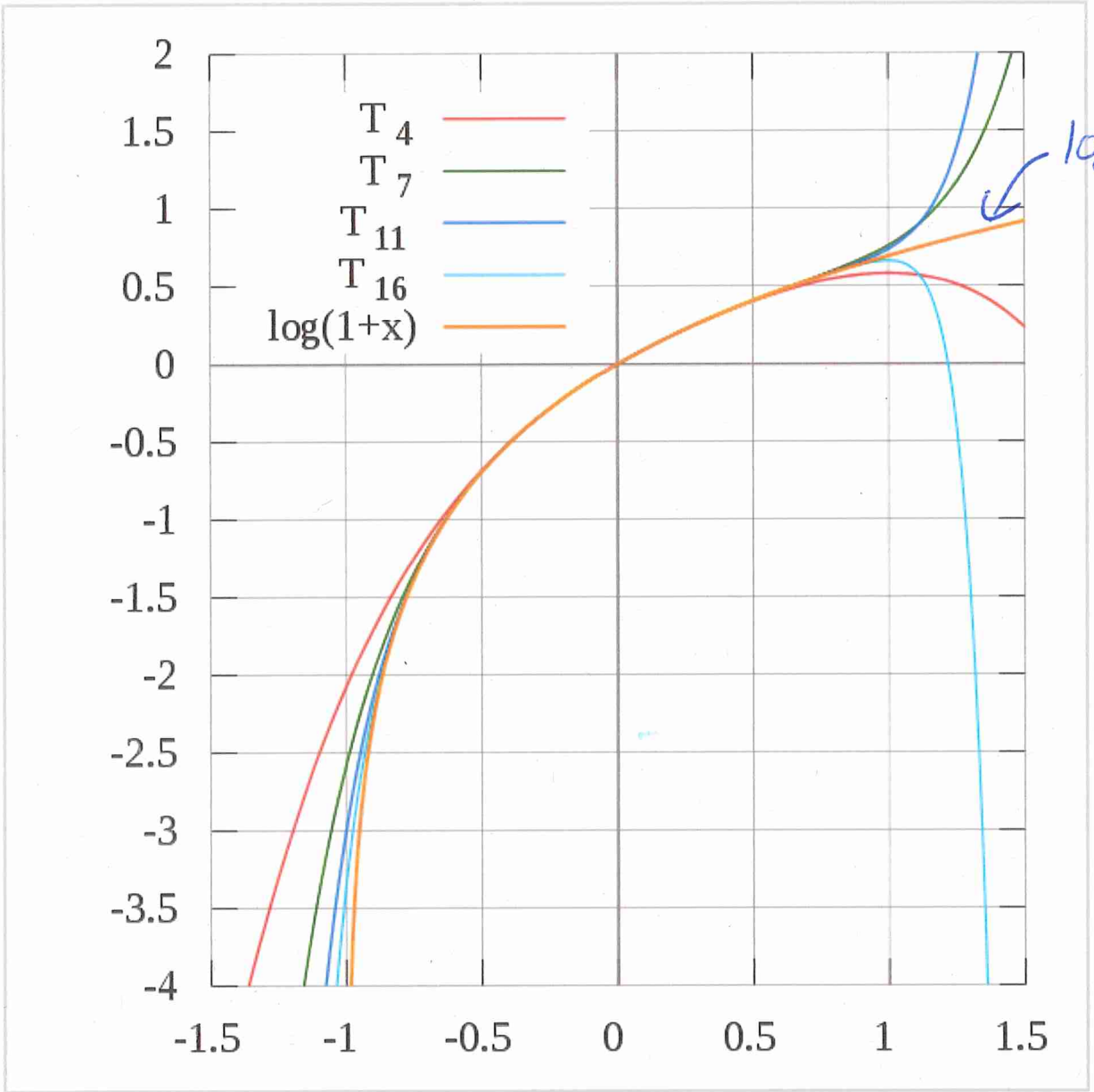


Taylor Series: sin(x)



As the degree of the Taylor polynomial rises, it approaches the correct function. This image shows $\sin x$ and its Taylor approximations, polynomials of degree 1, 3, 5, 7, 9, 11 and 13.

From Wikepedia (Accessed Feb. 25, 2019)



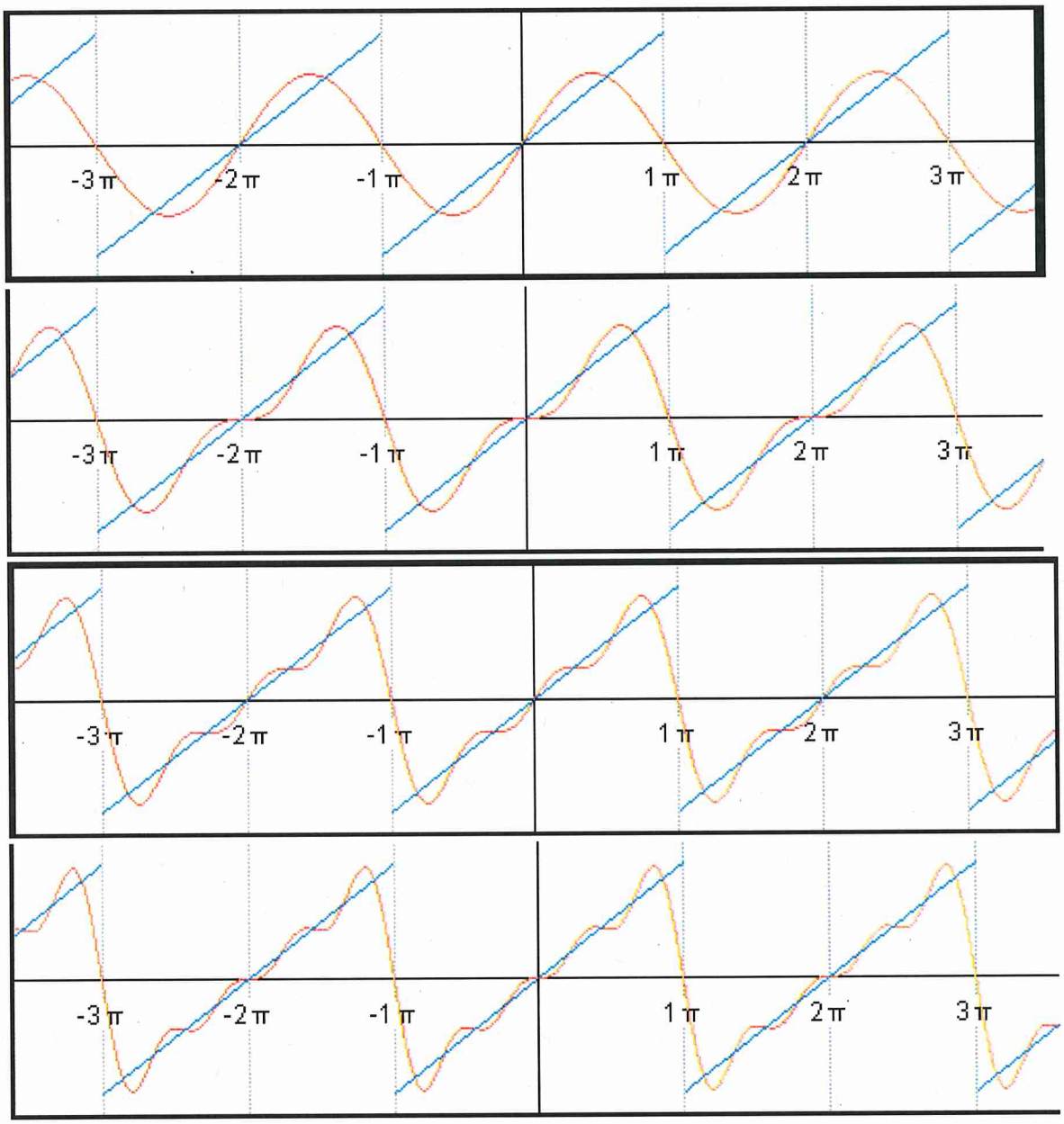
The Taylor polynomials for $\log(1+x)$ only provide accurate approximations in the range $-1 < x \leq 1$. For $x > 1$, Taylor polynomials of higher degree provide worse approximations.

From Wikipedia, accessed Feb. 26, 2019

Fourier Series Approximation – Wikipedia

Blue line – function we are approximating – linear.

One term, 2 terms, 4 terms, 5 terms



How do we calculate the coefficients?

We will use the following properties: (HW)

p. 207 Z+T

$$\int_{-\pi}^{\pi} \cos(nx)\cos(kx)\,dx = 0 \quad k \neq n$$

$$\int_{-\pi}^{\pi} \sin(nx)\sin(kx)\,dx = 0 \quad k \neq n$$

$$\int_{-\pi}^{\pi} \cos(nx)\sin(kx)\,dx = 0 \quad k = n, + k \neq n$$

$$\int_{-\pi}^{\pi} \cos^2(nx)\,dx = \int_{-\pi}^{\pi} \sin^2(nx)\,dx = \pi$$

$$\int_{-\pi}^{\pi} 1\,dx = 2\pi$$

Recall from linear algebra $\{\underline{v}_1, \underline{v}_2, \underline{v}_3\}$ basis vectors forms an orthogonal

$$\underline{v}_1 \cdot \underline{v}_2 = \underline{v}_1 \cdot \underline{v}_3 = \underline{v}_2 \cdot \underline{v}_3 = 0$$

$$\underline{v}_1 \cdot \underline{v}_1 \neq 0 \quad \underline{v}_2 \cdot \underline{v}_2 \neq 0 \quad \underline{v}_3 \cdot \underline{v}_3 \neq 0$$

$$f(\theta) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos(n\theta) + b_n \sin(n\theta) \right]$$

~~Multibl~~ Multiply both sides by $\cos(m\theta)$

& Then ~~#~~ integrate $\int_{-\pi}^{\pi}$

$$\int_{-\pi}^{\pi} f(\theta)\cos(m\theta)\,d\theta = \overbrace{\int_{-\pi}^{\pi} \frac{a_0}{2}\cos(m\theta)\,d\theta}^{=0}$$

$$+ \sum_{n=1}^{\infty} \left[\int_{-\pi}^{\pi} a_n \cos(n\theta)\cos(m\theta)\,d\theta \right\} \begin{array}{l} =0 \text{ except} \\ \text{when } n=m \\ =\pi a_m \end{array}$$

$$+ \int_{-\pi}^{\pi} b_n \sin(n\theta)\cos(m\theta)\,d\theta \Big] = 0$$

$$\int_{-\pi}^{\pi} f(\theta)\cos(m\theta)\,d\theta = a_m \pi \qquad a_m = \frac{1}{\pi}\int_{-\pi}^{\pi} f(\theta)\cos(m\theta)\,d\theta$$

$m = 1, 2, 3, \ldots$

(HW) $b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(\theta)\sin(n\theta)\,d\theta$

Let $n=0$ $\quad a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(\theta)\,d\theta$

Warning!

$$a_0 \neq \frac{1}{\pi}\int_{-\pi}^{\pi} f(\theta)\cos(n\theta)\,d\theta$$ (integrating 1st)

and then setting $n=0$.

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(\theta)\,d\theta$$ set $n=0$ first, then integrate.

ex) $f(\theta) = f_0$ is a constant.

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f_0 \cos(n\theta)\,d\theta = \frac{1}{\pi}\frac{1}{n} f_0 \sin(n\theta)\Big|_{\theta=-\pi}^{\pi} = 0$$

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f_0\,d\theta = \frac{f_0}{\pi}(2\pi) = 2f_0$$

$$b_n = 0$$

$$\text{F.S. } f(\theta) = \frac{a_0}{2} + \underbrace{\ldots}_{0} = \frac{2f_0}{2} = f_0$$

So for full F-S, need 3 integrals: one for

F-S $f(\theta) = \frac{a_0}{2} + \sum_{n=1}^{\infty} [a_n \cos(n\theta) + b_n \sin(n\theta)]$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(\theta)\cos(n\theta)\,d\theta$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(\theta)\sin(n\theta)\,d\theta$$

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(\theta)\,d\theta$$

① F-S $f(\theta)$ is periodic $-\pi < \theta < \pi$

② What f'ns have a F-S? (every term is finite):

Ans: when all integrals ~~are~~ for a_0, a_n, b_n are finite.

③ When does the F-S of a fn converge to the fn?

Ans: If $f(\theta)$ is <u>piecewise smooth</u> on the interval $-\pi \le \theta \le \pi$, then the F-S of $f(\theta)$ converges to

<u>Def'n</u> A fn, $f(x)$, is piecewise smooth (over some interval) if the ~~t~~ interval can be broken up into pieces (or sections) such that in each piece, $f(x)$ is C^1.